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Device, method, and non-transitory computer-readable medium for preparing single-line diagram

Abstract

A device generating a single-line diagram of each power system by diverting a part of a configuration common to the power systems is provided when a plurality of power systems having similar configurations exist. The device includes an input portion, a controller, and a storage. The input portion receives first setting related to a first power system, second setting related to a second power system, and third setting common to the first and second power systems. The controller generates setting data of a first single-line diagram including the first setting and the third setting, generates setting data of a second single-line diagram including the second setting and the third setting, and stores the setting data of the first single-line diagram and the setting data of the second single-line diagram in the storage.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a technique for preparing a single-line diagram, and more specifically, to reduction of man-hours for preparing the single-line diagram.

BACKGROUND ART

(2) Regarding the single-line diagram, for example, Japanese Patent Laying-Open No. 2002-34152 (PTL 1) discloses a substation facility management device in which “an actual instrument data preparing portion prepares actual instrument data of a single-line diagram, a protection linkage, and an operation linkage to which instrument individual information is written by writing the instrument individual information of the search result in a corresponding region in each design data based on a search result obtained by transmitting a search command to each search unit and previously-retaining design data of the single-line diagram, design data of the protection linkage, and design data of the operation linkage, and outputs a creation result” (see [abstract]).

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Patent Laying-Open No. 2002-34152

SUMMARY OF THE INVENTION

Technical Problem

(4) According to the technique disclosed in PTL 1, when a plurality of similar power systems exist, the single-line diagram of each power system is required to be individually prepared even when most of the configurations of the power systems are the same. Accordingly, when the plurality of power systems having similar configurations exist, a technique for preparing the single-line diagram of each power system is required by diverting a part of a configuration common to each power system.

(5) The present disclosure has been made in view of the background as described above, and an object in one aspect is to provide a technique for preparing the single-line diagram of each power system by diverting the part of the configuration common to each power system when the plurality of power systems having similar configurations exist.

Solution to Problem

(6) A device according to an embodiment includes an input portion to receive an operation related to a single-line diagram, a controller to generate setting data of the single-line diagram based on the operation, and a storage to store the setting data. The input portion receives first setting related to a first power system, second setting related to a second power system, and third setting common to the first and second power systems. The controller generates setting data of a first single-line diagram including the first setting and the third setting, generates setting data of a second single-line diagram including the second setting and the third setting, and stores the setting data of the first single-line diagram and the setting data of the second single-line diagram in the storage.

Advantageous Effects of Invention

(7) According to the embodiment, when the plurality of power systems having similar configurations exist, the single-line diagram of each power system can be prepared by diverting the part of the configuration common to each power system.

(8) The foregoing and other objects, features, aspects and advantages of the present disclosure content will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view illustrating an example of display of a single-line diagram **100** according to an embodiment.

- (2) FIG. 2 is a view illustrating an example of single-line diagrams **200A**, **200B** of power systems A, B, respectively.
- (3) FIG. 3 is a view illustrating an example of data related to single-line diagrams **300A**, **300B** including common setting and individual setting.
- (4) FIG. 4 is a view illustrating an example of data generated by a setting device **400** according to an embodiment.
- (5) FIG. 5 is a view illustrating an example of a circumstance of generation of setting data in a single-line diagram of each monitoring device.
- (6) FIG. 6 is a view illustrating an example of a relationship between the single-line diagram displayed by each monitoring device and a setting file of each monitoring device.
- (7) FIG. 7 is a view illustrating an example of a layer included in the setting file.
- (8) FIG. 8 illustrates an example of a configuration of a functional block of the setting device **400**.
- (9) FIG. 9 illustrates an example of a hardware configuration of an information processing device **900**.
- (10) FIG. 10 is a view illustrating an example of editing data of the single-line diagram and setting data of the single-line diagram that are stored in the setting device **400**.
- (11) FIG. 11 is a view illustrating an example of a symbol included in the layer.
- (12) FIG. 12 is a view illustrating an example of a flowchart in internal processing of the setting device **400**.
- (13) FIG. 13 is a view illustrating an example of a setting file **1300** according to an embodiment.
- (14) FIG. 14 illustrates an example of a configuration of a functional block in a setting device **1400** according to an embodiment.
- (15) FIG. 15 is a view illustrating an example of the editing data of a single-line diagram and the setting data of the single-line diagram that are stored in the setting device **1400**.
- (16) FIG. 16 is a view illustrating an example of incidental information **1600** about internal data **1310**.
- (17) FIG. 17 is a view illustrating an example of a configuration of a setting file **1700** included in the editing data of the single-line diagram.
- (18) FIG. 18 is a view illustrating a first example of a screen of an editing screen **1810** provided by an editing portion **1401**.
- (19) FIG. 19 is a view illustrating a second example of the screen of the editing screen **1810** provided by the editing portion **1401**.
- (20) FIG. 20 is a view illustrating a third example of the screen of the editing screen **1810** provided by the editing portion **1401**.

DESCRIPTION OF EMBODIMENTS

(21) Hereinafter, an embodiment of the technical idea according to the present disclosure will be described with reference to the drawings. In the following description, the same component is denoted by the same reference numeral. Those names and functions are the same. Accordingly, the detailed description thereof will not be repeated. In the following description, when referring to a plurality of the same configurations, they are sometimes expressed as configurations **123A**, **123B**. In addition, the plurality of the same configurations are expressed as a configuration **123** when collectively referred to.

First Embodiment

(22) FIG. 1 is a view illustrating an example of display of a single-line diagram **100** according to a first embodiment. The “single-line diagram” displays a state of a power system and supplementary information using an image, a numerical value, a character, and the like. Single-line diagram **100** is displayed on a display or the like of a monitoring device. For example, the “monitoring device” monitors a certain facility in a substation. An administrator of the power system can grasp the state of the power system and appropriately perform maintenance of the facility by checking single-line diagram **100**.

- (23) Single-line diagram **100** represents a certain power system, and roughly includes a symbol **110** and internal data **120**. Single-line diagram **100** can be prepared by the number of power systems of a maintenance target. For example, when three power systems of the maintenance target exist, single-line diagram **100** corresponding to each of the three power systems of the maintenance target is prepared.
- (24) Symbol **110** indicates a configuration included in the power system. In one aspect, the configuration can include a component, a facility, or an instrument, such as a power line and a switch, which constitute the power system. In another aspect, symbol **110** may further include components such as a label, a text box, and a button, which display the supplementary information of the power system.
- (25) Internal data **120** is dynamically changing data associated with each symbol. The “internal data” represents an actual measurement value of information about each symbol. For example, the information about the symbol is voltage or current. In other words, the internal data is data indicating the internal state of a certain facility or instrument. The numerical value displayed in the text box or the like can dynamically change according to the state of internal data **120**. As an example, internal data **120** in FIG. 1 includes the voltage at an X-bus and the voltage at a Y-bus. Internal data **120** dynamically changes according to the state of internal data **120** including the voltage at the X-bus and the voltage at the Y-bus of a certain facility. In one aspect, internal data **120** can represent the voltage, the current, any information obtained from an arithmetic operation, and any sensor values (gas pressure, temperature, and the like).
- (26) The monitoring device displays single-line diagram **100** on the display by reading setting data of the single-line diagram. For example, the “setting data of the single-line diagram” includes the number of symbols **110**, coordinates, color information, and the like included in the single-line diagram and association information about each symbol **110** and internal data **120**.
- (27) The setting data of the single-line diagram is generated in the setting device of the single-line diagram. The “setting device in the single-line diagram (hereinafter, the setting device in the single-line diagram is simply referred to as a “setting device”)” has a function for receiving operation input from a user and a function for editing a screen of single-line diagram **100**, associating each symbol **110** and each internal data **120**, and the like based on the operation input. In one aspect, setting device **400** may be a personal computer, a workstation, or a virtual machine built on a cloud environment.
- (28) In another aspect, the setting device and the monitoring device may be the same instrument. In another aspect, the setting device and the monitoring device may be different instruments. In this case, the monitoring device can display single-line diagram **100** on the display by acquiring the setting data of the single-line diagram from the setting device.
- (29) With reference to FIGS. 2 and 3, a problem in preparing a plurality of single-line diagrams will be described below. Hereinafter, it is assumed that a certain substation includes a plurality of power systems A, B (individual facilities A, B).
- (30) FIG. 2 is a view illustrating an example of single-line diagrams **200A**, **200B** of power systems A, B, respectively. Single-line diagram **200A** includes a symbol **210A** and a symbol **220A**. Single-line diagram **200B** includes a symbol **210B** and a symbol **220B**.
- (31) When the single-line diagram displayed on each of the plurality of monitoring devices installed in one substation is a single-line diagram of a power system connected to the same bus, the single-line diagrams often have configurations similar to each other as illustrated in FIG. 2. However, although the single-line diagrams have many common setting contents, the single-line diagrams are not completely the same, and only a part of the numbers of the instruments or the names of the buses may be different. For example, in single-line diagrams **200A**, **200B**, symbol **210A** and symbol **210B** are common. However, it can be seen that symbol **220A** and symbol **220B** have different bus names.
- (32) In the conventional method for preparing the single-line diagram, the user need to individually

prepare single-line diagrams **200A**, **200B** having only different details as described above. Such single-line diagrams **200A**, **200B** include the common setting and the individual setting.

(33) FIG. **3** is a view illustrating an example of data related to single-line diagrams **300A**, **300B** including the common setting and the individual setting. With reference to FIG. **3**, the data that causes an increase in man-hours for preparing the plurality of single-line diagrams are prepared will be described. Single-line diagram **300A** is displayed on the screen of monitoring device A, and single-line diagram **300B** is displayed on the screen of monitoring device B.

(34) The user prepares editing data of the single-line diagram using the setting device. The “editing data of the single-line diagram” is data that can be edited by the setting device. The user can edit the single-line diagram by editing the editing data of the single-line diagram. The setting device generates the setting data of the single-line diagram based on the editing data of the single-line diagram. The monitoring device displays the single-line diagram on the display based on the setting data of the single-line diagram.

(35) In the example of FIG. **3**, the user edits editing data **301A** of the single-line diagram using the setting device. The setting device generates setting data **303A** of the single-line diagram read by monitoring device A based on editing data **301A** of the single-line diagram. Monitoring device A displays single-line diagram **300A** on the display based on setting data **303A** of the single-line diagram.

(36) Similarly, the user edits editing data **301B** of the single-line diagram using the setting device. The setting device generates setting data **303B** of the single-line diagram read by monitoring device B based on editing data **301B** of the single-line diagram. Monitoring device B displays single-line diagram **300B** on the display based on setting data **303B** of the single-line diagram.

(37) Editing data **301A**, **301B** of the single-line diagram include a common setting **310** and an individual setting **320** for each of monitoring devices A, B. In the conventional technique, the user needs to prepare common setting **310** and individual setting **320** for both editing data **301A**, **301B** of the single-line diagram. This causes the increase in the man-hours for preparing the single-line diagram. Accordingly, the setting device of the first embodiment provides a function that can use common setting **310** when editing data **301A**, **301B** of the single-line diagram are prepared.

(38) With reference to FIGS. **4** to **7**, the function provided by the setting device of the first embodiment will be described below. The setting device of the first embodiment provides the function for preparing each of the single-line diagrams by diverting the common setting to each of the single-line diagrams during the preparation of the plurality of single-line diagrams. Details of the function will be described below.

(39) FIG. **4** is a view illustrating an example of data generated by a setting device **400** of the first embodiment. It is assumed that setting device **400** generates the setting data of the single-line diagram for three monitoring devices A, B, C.

(40) Setting device **400** generates editing data **410** of the single-line diagram based on the operation input of the user. Editing data **410** of the single-line diagram includes a common setting **411** and an individual setting **412**. The “common setting” is a setting common to each monitoring device. The “individual setting” is an individual setting of each monitoring device. As an example, symbols common to the single-line diagrams displayed by monitoring devices A, B, C are included in common setting **411**. The symbol included only in the single-line diagram displayed by monitoring device A is included in individual setting **412A**. Hereinafter, when the common setting and the individual setting are collectively referred to, they are simply referred to as “setting”.

(41) Common setting **411** and individual setting **412** include a layer. Alternatively, common setting **411** and individual setting **412** themselves may be the layer. The “layer” is obtained by disposing one or a plurality of symbols in a certain region. Setting device **400** can superimpose these layers. For example, the layer of common setting **411** includes one or the plurality of symbols common to the single-line diagrams displayed by monitoring devices A, B, C. The layer of individual setting **412** includes one or the plurality of symbols included only in the single-line diagram displayed by

each of monitoring devices A, B, C.

(42) A setting data output portion **415** generates single-line diagram setting data by superimposing the layer of common setting **411** and the layer of individual setting **412**. As an example, setting data output portion **415** generates the setting data of the single-line diagram for monitoring device A by superimposing the layer of common setting **411** and the layer of individual setting **412A**. In one aspect, setting data output portion **415** can arbitrarily change overlapping order between the layer of common setting **411** and the layer of individual setting **412** A based on operation input of the user. In another aspect, setting data output portion **415** can change display/non-display setting of each layer or each symbol included in each layer based on the operation input of the user.

(43) Setting data output portion **415** stores setting data of the single-line diagrams for monitoring devices A to C in databases (DBs) **420A** to **420C** of the monitoring devices. Monitoring devices A to C can display the single-line diagrams on the display based on the setting data of the single-line diagrams acquired from DBs **420A** to **420C** of the respective monitoring devices. In one aspect, DBs **420A** to **420C** of the monitoring devices may be a single DB. In another aspect, setting data output portion **415** may only write the setting data of the single-line diagrams for monitoring devices A to C to a nonvolatile memory.

(44) FIG. 5 is a view illustrating an example of a circumstance of generation of setting data in the single-line diagram of each monitoring device. The setting data of the single-line diagram for monitoring device A is generated based on common setting **411** and individual setting **412A**. The setting data of the single-line diagram for monitoring device B is generated based on common setting **411** and individual setting **412B**. It can be seen that common setting **411** is used not only when the setting data of the single-line diagram of monitoring device A is generated but also when the setting data of the single-line diagram for monitoring device B is generated.

(45) FIG. 6 is a view illustrating an example of a relationship between the single-line diagram displayed by each monitoring device and a setting file of each monitoring device. At this point, the “setting file” refers to a group of data generating the setting data of the single-line diagram. For example, the setting file of monitoring device A includes common setting **411** and individual setting **412A**. Alternatively, when simply referred to as the setting file, the setting file can include common setting **411** and individual settings **412** of all the monitoring devices.

(46) Referring to FIG. 6, a setting file **600A** for monitoring device A includes common setting **411** and individual setting **412A** for monitoring device A. A setting file **600B** for monitoring device B includes common setting **411** and individual setting **412B** for monitoring device B.

(47) Monitoring device A displays a single-line diagram **601A** on the display based on the single-line diagram for monitoring device A. The symbol common to each monitoring device included in single-line diagram **601A** is included in common setting **411**. The symbol individual for monitoring device A included in single-line diagram **601A** are included in individual setting **412A**.

(48) Similarly, monitoring device B displays a single-line diagram **601B** on the display based on the single-line diagram for monitoring device B. The symbol common to the respective monitoring devices included in single-line diagram **601B** are included in common setting **411**. The symbol individual for monitoring device B included in single-line diagram **601B** are included in individual setting **412B**.

(49) FIG. 7 is a view illustrating an example of the layer included in the setting file. A setting file **700** in FIG. 7 includes common setting **411**, individual setting **412A**, and individual setting **412B**. In other words, setting file **700** includes the common layer, the layer for monitoring device A, and the layer for monitoring device B. Setting file **700** may also include information on overlapping order of these layers. When generating the setting data of the single-line diagram of each monitoring device, setting device **400** can combine the settings (common setting **411** and individual setting **412**) based on the information about the overlapping order of the layers.

(50) As described with reference to FIGS. 4 to 7, setting device **400** has the function for generating the setting data of the single-line diagram for each monitoring device using common setting **411**

common to each monitoring device. With this function, the user can divert common setting **411** to easily prepare the setting data of the single-line diagram for each monitoring device by preparing common setting **411** common to each monitoring device only once. As a result, the man-hours for preparing the setting data of the single-line diagram can be reduced.

(51) With reference to FIGS. **8** to **12**, the details of the software and hardware configuration of setting device **400** and the internal operation of setting device **400** will be described below.

(52) FIG. **8** illustrates an example of a configuration of a function for implementing setting device **400**. In one aspect, the functional block in FIG. **8** can be implemented as hardware. In another aspect, the functional block in FIG. **8** can be implemented as software. In that case, each functional block may be executed on the hardware in FIG. **9**. Setting device **400** includes an editing portion **801**, a layer processing portion **802**, an editing data storage **803**, a setting data output portion **804**, and a setting data storage **805**.

(53) Editing portion **801** provides an editing screen (user interface) editing the editing data of the single-line diagram. The user can execute the preparation, editing, deletion, and the like of the editing data of the single-line diagram by operating the editing screen from the input device provided in setting device **400**. When receiving a storage operation, editing portion **801** stores the editing data of the single-line diagram in editing data storage **803** through layer processing portion **802**.

(54) Layer processing portion **802** executes processing related to each layer included in the editing data of the single-line diagram. For example, layer processing portion **802** provides a layer-by-layer display/non-display switching function. In addition, layer processing portion **802** executes processing for superimposing each layer or the like based on the setting of superimposition order input by the user. Layer processing portion **802** superimposes and displays each layer included in the editing data of the single-line diagram on the editing screen based on the setting of the overlapping order of the layers.

(55) In one aspect, layer processing portion **802** can superimpose each layer to generate editable editing data of the single-line diagram, and output the editing data of the single-line diagram to editing portion **801**. In another aspect, layer processing portion **802** may be incorporated in editing portion **801** as one function. When the storage processing is executed in editing portion **801**, layer processing portion **802** stores the editing data of the single-line diagram in editing data storage **803**.

(56) Setting device **400** can provide various functions by combining editing portion **801** and layer processing portion **802**. For example, setting device **400** can switch between the display and the non-display in units of layers or symbols based on the operation input on the editing screen. When a certain layer becomes the non-display, the symbols disposed on the layer becomes also the non-display. In addition, setting device **400** can change the overlapping order of the layers based on the operation input on the editing screen. Furthermore, setting device **400** can dispose the symbol in each layer based on the operation input on the editing screen.

(57) Editing data storage **803** stores the editing data of the single-line diagram. In one aspect, editing data storage **803** may store the editing data of individual single-line diagrams. In another aspect, editing data storage **803** may store each layer and reference information for each layer. In this case, editing portion **801**/layer processing portion **802** can reconstruct the editing data of each single-line diagram from each layer and the reference information for each layer based on the reception of the operation of reading the editing data of the single-line diagram.

(58) Setting data output portion **804** converts the editing data of the single-line diagram stored in editing data storage **803** into the setting data of the single-line diagram, and stores the setting data of the single-line diagram in setting data storage **805**. Each setting includes “application target information” indicating which monitoring device the setting itself is for. For example, the common setting includes the application target information indicating that the common setting is common to all the monitoring devices. The individual setting of monitoring device A includes the application target information indicating that the data is for monitoring device A. Setting data output portion

804 can generate the setting data of the single-line diagram by combining required settings based on these pieces of application target information.

(59) Setting data storage **805** stores the setting data of the single-line diagram. In one aspect, setting data storage **805** may transmit the setting data of the single-line diagram to the monitoring device based on acceptance of a request for the setting data of the single-line diagram from the monitoring device.

(60) FIG. **9** illustrates an example of a hardware configuration of information processing device **900**. Setting device **400** can be implemented by information processing device **900**. Information processing device **900** includes a central processing unit (CPU) **901**, a primary storage device **902**, a secondary storage device **903**, an external instrument interface **904**, an input interface **905**, an output interface **906**, and a communication interface **907**. In one aspect, setting device **400** may be implemented by combining a plurality of information processing devices **900**.

(61) CPU **901** can execute a program implementing various functions of information processing device **900**. For example, CPU **901** is constructed with at least one integrated circuit. For example, the integrated circuit may be constructed of at least one CPU, at least one field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a combination thereof, or the like.

(62) Primary storage device **902** stores the program executed by CPU **901** and data referred to by CPU **901**. In one aspect, primary storage device **902** may be implemented by a dynamic random access memory (DRAM), a static random access memory (SRAM), or the like.

(63) Secondary storage device **903** is a nonvolatile memory, and may store the program executed by CPU **901** and the data referred to by CPU **901**. In this case, CPU **901** executes the program read from secondary storage device **903** to primary storage device **902**, and refers to the data read from secondary storage device **903** to primary storage device **902**. In one aspect, secondary storage device **903** may be implemented by a hard disk drive (HDD), a solid state drive (SSD), an erasable programmable read only memory (EPROM), an electrically erasable programmable read only memory (EEPROM), a flash memory, or the like.

(64) External instrument interface **904** can be connected to any external instrument such as a printer, a scanner, and an external HDD. In one aspect, external instrument interface **904** may be implemented by a universal serial bus (USB) terminal or the like.

(65) Input interface **905** can be connected to any input device such as a keyboard, a mouse, a touchpad, or a game pad. In one aspect, input interface **905** may be implemented by a USB terminal, a PS/2 terminal, a Bluetooth (registered trademark) module, and the like.

(66) Output interface **906** can be connected to any output device such as a cathode ray tube display, a liquid crystal display, or an organic electro-luminescence (EL) display. In one aspect, output interface **906** may be implemented by a USB terminal, a D-sub terminal, a digital visual interface (DVI) terminal, a high-definition multimedia interface (HDMI) (registered trademark) terminal, or the like.

(67) Communication interface **907** is connected to a wired or wireless network instrument. In one aspect, communication interface **907** may be implemented by a wired local area network (LAN) port, a wireless fidelity (Wi-Fi (registered trademark)) module, or the like. In another aspect, communication interface **907** may transmit and receive the data using a communication protocol such as Transmission Control Protocol/Internet Protocol (Transmission Control Protocol/Internet Protocol) or User Datagram Protocol (UDP).

(68) FIG. **10** is a view illustrating an example of the editing data of the single-line diagram and the setting data of the single-line diagram that are stored in setting device **400**. The editing data of the single-line diagram can be stored in secondary storage device **903**. Secondary storage device **903** stores editing data **1000** of the single-line diagram and setting data **1030** of the single-line diagram.

(69) Editing data **1000** of the single-line diagram is stored in a region secured as editing data storage **803** on secondary storage device **903**. Editing data storage **803** can store editing data **1000** of one or the plurality of single-line diagrams. Editing data **1000** of the single-line diagram includes

one or the plurality of settings (common setting or/and individual setting).

(70) Each setting includes one layer in editing data **1000** of the single-line diagram. Furthermore, each setting includes the setting of the overlapping order of layers, the setting of display/non-display of layers, and the setting of display/non-display of symbols included in layers. In the example of FIG. **10**, editing data **1000** of the single-line diagram includes common setting **1010**, individual setting **1020A** of monitoring device A, and individual setting **1020B** of monitoring device B. Common setting **1010** includes the layer common to monitoring devices A, B, and each of individual settings **1020A**, **1020B** includes the individual layer for monitoring devices A, B.

(71) The layer can constitute by combining one or a plurality of pre-componentized symbols. Each symbol is a partial element constituting the single-line diagram, and may be a figure component (for example, a square, a line, a circle, or the like), a character string, a numerical value, any other component, or a combination thereof.

(72) In addition, the layer includes a region in which the symbol can be disposed. A size of the region is common in the layers. Furthermore, the layer includes application target information. The application target information indicates whether the layer is the layer common among all the monitoring devices (the layer of the common setting) or the individual layer of a certain monitoring device (the layer of the individual setting).

(73) Setting data **1030** of the single-line diagram is stored in the region secured as setting data storage **805** on secondary storage device **903**. In the example of FIG. **10**, setting data **1030A** of the single-line diagram for monitoring device A and setting data **1030B** of the single-line diagram for monitoring device B are stored in secondary storage device **903**.

(74) FIG. **11** is a view illustrating an example of the symbol included in the layer. In the example of FIG. **11**, setting device **400** stores editing data **1110A** of the single-line diagram and editing data **1110B** of the single-line diagram.

(75) It is assumed that setting device **400** stores the editing data of the single-line diagram including layers 1 to 3 (setting). The user can easily produce editing data (corresponding to setting files **600A**, **600B** in FIG. **6**) of the single-line diagram for each monitoring device using layers 1 to 3 by preparing layers 1 to 3 only once.

(76) For example, setting device **400** can prepare editing data **1110A** (the setting file of monitoring device A) of the single-line diagram for monitoring device A based on the operation input of the user. Similarly, setting device **400** can prepare editing data **1110B** (the setting file of monitoring device B) of the single-line diagram for monitoring device B based on the operation input of the user. Editing data **1110A** of the single-line diagram and editing data **1110B** of the single-line diagram may include only the reference information about layers 1 to 3, the setting of the overlapping order, and the setting of the display/non-display.

(77) The setting of the layer overlapping order of editing data **1110A** in the single-line diagram is in the order of layer 1, layer 2, and layer 3. In addition, layers 3, 2 are set to be displayed, and layer 1 is set to be non-displayed. For this reason, the display of editing data **1110A** of the single-line diagram on the display is as illustrated in a display **1120A**.

(78) In addition, the setting of the layer overlapping order of editing data **1110B** of the single-line diagram is in the order of layer 2, layer 1, and layer 3. In addition, layers 1, 2 are set to be displayed, and layer 3 is set to be non-displayed. For this reason, the display of editing data **1110B** of the single-line diagram on the display is as illustrated in a display **1120B**.

(79) FIG. **12** is a view illustrating an example of a flowchart of internal processing in setting device **400**. In one aspect, CPU **901** may read the program performing the processing in FIG. **12** from secondary storage device **903** to primary storage device **902** and execute the program. In another aspect, a part or all of the processing can be implemented as a combination of circuit elements configured to execute the processing.

(80) The flowchart in FIG. **12** illustrates a procedure of processing for generating the setting data of the single-line diagram from the editing data of the single-line diagram.

(81) In step **S1210**, CPU **901** acquires the editing data of the single-line diagram from secondary storage device **903**. In one aspect, the editing data of the single-line diagram acquired by CPU **901** in this step may be the editing data of the single-line diagram including only the reference information about the layer, the setting of the overlapping order, and the setting of the display/non-display (corresponding to editing data **1110A**, **1110B** of the single-line diagram in FIG. **11**).

(82) In step **S1220**, CPU **901** specifies the monitoring device that outputs the setting data of the single-line diagram from the application target information about the layer included in the acquired editing data of the single-line diagram. For example, when the application target information about the layer included in the acquired editing data of the single-line diagram indicates monitoring device A, CPU **901** specifies monitoring device A as an output destination of the setting data of the single-line diagram. CPU **901** may specify the plurality of monitoring devices as the output destinations of the setting data of the single-line diagram.

(83) In step **S1230**, CPU **901** selects one of the monitoring devices specified in step **S1220** as the output destination of the setting data of the single-line diagram. In one aspect, when specifying monitoring devices A to C in step **S1220**, CPU **901** can automatically select any one of monitoring devices A to C. In another aspect, CPU **901** may determine the monitoring device selected by the input operation as the output destination of the setting data of the single-line diagram based on the reception of the input operation selecting any one of the monitoring devices specified in step **S1220** through input interface **905**.

(84) In step **S1240**, CPU **901** acquires the layer output to the monitoring device selected in step **S1230**. For example, when selecting monitoring device A in step **S1230**, CPU **901** selects the layer including the application target information indicating monitoring device A/all the monitoring devices. In one aspect, CPU **901** may acquire the layer based on the reference information about the layer acquired in step **S1210**.

(85) In step **S1250**, CPU **901** determines whether the layer acquired in step **S1240** is the individual layer for a certain monitoring device or the layer common to all the monitoring devices. When determining that the layer acquired in step **S1240** is the individual layer for a certain monitoring device or the layer common to all the monitoring devices (YES in step **S1250**), CPU **901** shifts the control to step **S1260**. Otherwise (NO in step **S1250**), CPU **901** shifts the control to step **S1270**.

(86) In step **S1260**, CPU **901** executes processing for outputting the layer acquired in step **S1240** to the setting data of the single-line diagram. For example, when monitoring device A is selected in step **S1230**, CPU **901** executes processing for adding the layer acquired in step **S1240** to the setting data of the single-line diagram of monitoring device A.

(87) In step **S1270**, CPU **901** determines whether the non-output layer for the setting data of the single-line diagram exists. For example, when selecting monitoring device A in step **S1230**, CPU **901** determines whether the layer that is not yet output to the setting data of the single-line diagram of monitoring device A exists among the layers to be added to the setting data of the single-line diagram of monitoring device A.

(88) When CPU **901** determines that the non-output layer for the setting data of the single-line diagram exists (YES in step **S1270**), the control proceeds to step **S1240**. Otherwise (NO in step **S1270**), CPU **901** shifts the control to step **S1280**.

(89) In step **S1280**, CPU **901** determines whether the monitoring device that does not output the setting data of the single-line diagram exists. For example, when specifying monitoring devices A to C in step **S1220**, CPU **901** determines whether the setting data of the single-line diagram is output to all monitoring devices A to C.

(90) When CPU **901** determines that the monitoring device that does not output the setting data of the single-line diagram exists (YES in step **S1280**), the control proceeds to step **S1230**. Otherwise (NO in step **S1280**), CPU **901** ends the processing.

(91) In one aspect, CPU **901** may directly output the setting data of the single-line diagram to each monitoring device. In another aspect, CPU **901** may store the setting data of the single-line diagram

in setting data storage **805**.

(92) As described above, setting device **400** of the first embodiment has the function for generating the editing data/setting data of the single-line diagram for each monitoring device by diverting common setting **411** common to each monitoring device. With this function, the user can divert common setting **411** to easily prepare the setting data of the single-line diagram for each monitoring device by preparing common setting **411** common to each monitoring device only once. As a result, the man-hours for preparing the setting data of the single-line diagram can be reduced.

(93) In addition, each setting includes the layer, and setting device **400** has the function for easily generating the setting data of the single-line diagram for each monitoring device based on the combination, the overlapping order, and the display/non-display settings of each layer. With the function, the user can update the editing data/setting data of all the single-line diagrams including a certain layer only by editing the symbol included in the certain layer. As a result, the maintainability of the setting data of the single-line diagram is also improved.

Second Embodiment

(94) A second embodiment will be described below. A setting device of the second embodiment is different from the setting device of the first embodiment in that each layer has a function for associating the internal data. When the single-line diagram displayed in each of a plurality of monitoring devices of the same type installed in one substation is a single-line diagram of a power system connected to the same bus, the single-line diagram may include not only the symbol common to each other but also the internal data common to each other. Therefore, the setting device of the second embodiment can provide a function for more easily generating the setting data of the single-line diagram for each monitoring device by allocating allocation information of the internal data to the common setting.

(95) FIG. **13** is a view illustrating an example of a setting file **1300** of the second embodiment. Setting file **1300** generated and stored by the setting device of the second embodiment includes common setting **1320**, individual setting **1330** for each monitoring device, and internal data **1310** associated with each symbol included in the layer (each setting). In the example of FIG. **13**, the setting device stores individual settings **1330A**, **1330B** for monitoring devices A, B.

(96) The monitoring device displaying the single-line diagram can update a numerical value displayed on the single-line diagram in real time based on the internal data acquired from the power system. For example, the internal data is associated with the individual symbol on the layer, and represents an actual measurement value of information about the symbol. For example, the internal data associated with the X-bus in FIG. **1** represents the actual measurement value of the X-bus.

(97) In the example of FIG. **13**, “numerical data 1” is associated with the symbol of the layer of monitoring device B. “Numerical data 2” is associated with the symbol of the layer of monitoring device A. “Numerical data 3” is associated with the symbol of the layer common to all the monitoring devices.

(98) The setting device generates the setting data of the single-line diagram for each monitoring device based on common setting **1320** (the layer common to all the monitoring devices), individual setting **1330A** (the layer of monitoring device A), individual setting **1330B** (the layer of monitoring device B), and the association information about each internal data. For example, the setting data of the single-line diagram stored in DB **420A** of monitoring device A includes association information between “numerical data 2” and the symbol on the “layer of monitoring device A” and association information between “numerical data 3” and the symbol on the “layer common to all the monitoring devices”. Similarly, the setting data of the single-line diagram stored in DB **420B** of monitoring device B includes association information between “numerical data 1” and the symbol on “the layer of the monitoring device B” and association information between “numerical data 3” and the symbol on “the layer common to all the monitoring devices”.

(99) FIG. **14** illustrates an example of a configuration of functional blocks in a setting device **1400** of the second embodiment. The functional blocks in FIG. **14** further includes a function related to

the internal data in addition to the function of the functional block in FIG. 8. In one aspect, the functional blocks in FIG. 14 can be implemented as hardware. In another aspect, the functional blocks in FIG. 14 can be implemented as software. In that case, each functional block can be executed on the hardware of information processing device 900 in FIG. 9.

(100) Setting device 1400 includes an editing portion 1401, a layer processing portion 1402, an editing data storage 1403, a setting data output portion 1404, a setting data storage 1405, and an internal data storage 1406. Editing portion 1401 provides the editing screen for editing the editing data of the single-line diagram. The user can execute the operation of associating the internal data for each symbol disposed in each layer through the editing screen.

(101) Internal data storage 1406 stores the internal data of the power system of the maintenance target. In one aspect, the internal data may be an address of a memory in which a value of a certain sensor or the like is written. For example, editing portion 1401 can display a list of the internal data read from internal data storage 1406, and present the list to the user.

(102) Layer processing portion 1402 can also manage the association information about the internal data for each symbol disposed in each layer in addition to information about each layer included in the editing data of the single-line diagram. Editing data storage 1403 stores the editing data of the single-line diagram. The editing data of the single-line diagram includes the association information about the internal data with respect to each symbol disposed in each layer in addition to the information about the layer.

(103) Setting data output portion 1404 converts the editing data of the single-line diagram stored in editing data storage 1403 into the setting data of the single-line diagram, and stores the setting data of the single-line diagram in setting data storage 1405. At that time, setting data output portion 1404 includes the association information about the internal data with respect to each symbol disposed in each layer in the setting data of the single-line diagram. For example, the setting data of the single-line diagram of monitoring device A may include the association information about the internal data with respect to the symbol on the layer of monitoring device A and the association information about the internal data with respect to the symbol on the layer common to all the monitoring devices.

(104) Setting data storage 1405 stores the setting data of the single-line diagram. The setting data of the single-line diagram includes the association information about the internal data with respect to each symbol disposed in each layer. In one aspect, setting data storage 1405 may transmit the setting data of the single-line diagram to the monitoring device based on the acceptance of the request for the setting data of the single-line diagram from the monitoring device.

(105) FIG. 15 is a view illustrating an example of the editing data of the single-line diagram and the setting data of the single-line diagram that are stored in setting device 1400. The editing data of the single-line diagram can be stored in secondary storage device 903. In the example of FIG. 15, secondary storage device 903 stores internal data 1310, editing data 1500 of the single-line diagram, and setting data 1530 of the single-line diagram.

(106) Internal data 1310 is stored in a region secured as internal data storage 1406 on secondary storage device 903. In one aspect, internal data 1310 may be a set of the plurality of pieces of internal data (such as “numerical data 1 to 3”). Internal data storage 1406 stores the actual measurement value (current, voltage, or the like) of each facility in the power system of the maintenance target. The actual measurement value of each facility can be updated as needed.

(107) Editing data 1500 of the single-line diagram is stored in the region secured as editing data storage 1403 on secondary storage device 903. Editing data storage 1403 can store editing data 1500 of one or the plurality of single-line diagrams. Editing data 1500 of the single-line diagram includes the association information for the symbol of internal data 1310 or a part or all of internal data 1310 associated with the symbol in addition to all the information included in editing data 1000 of the single-line diagram in FIG. 10.

(108) For example, common setting 1320 includes association information 1540 about “internal

data 1” with respect to the symbol on the layer of common setting **1320**. Individual setting **1330A** of monitoring device A includes association information **1550A** about “internal data 2” with respect to the symbol on the layer of the individual setting **1330A**. Further, individual setting **1330B** of monitoring device B includes association information **1550B** about “internal data 3” with respect to the symbol on the layer of individual setting **1330B**.

(109) Setting data **1530** of the single-line diagram is stored in the region secured as setting data storage **1405** on secondary storage device **903**. Setting data storage **1405** can store setting data **1530** of one or the plurality of single-line diagrams. Setting data **1530** of the single-line diagram includes the association information for the symbol of internal data **1310** or a part or all of internal data **1310** associated with the symbol in addition to all the information included in setting data **1030** of the single-line diagram in FIG. **10**. For example, setting data **1530A** of the single-line diagram of monitoring device A includes association information **1540** about “internal data 1” and association information **1550A** of “internal data 2”. Setting data **1530B** of the single-line diagram of monitoring device B includes association information **1540** about “internal data 1” and the association information **1550B** about “internal data 3”.

(110) FIG. **16** is a view illustrating an example of incidental information **1600** about internal data **1310**. Internal data **1310** may include incidental information **1600** in addition to the actual measurement value of the facility in the power system of the maintenance target. As an example, incidental information **1600** may include an identifier (ID) **1601**, a unit **1602**, a data type **1603**, a maximum value **1604**, and a minimum value **1605**. In one aspect, setting device **1400** may store incidental information **1600** in the region of internal data storage **1406** in secondary storage device **903**. In another aspect, setting device **1400** may store incidental information **1600** in the region different from internal data storage **1406** in secondary storage device **903**. Setting device **1400** can appropriately refer to incidental information **1600**.

(111) ID **1601** uniquely identifies the actual measurement value (voltage, current, or the like) of each facility. Unit **1602** indicates the unit of the actual measurement value of each facility. Data type **1603** defines the data type of the actual measurement value of each facility. Maximum value **1604** is the maximum value of the actual measurement value of each facility. In one aspect, maximum value **1604** may be the maximum value of the actual measurement value measured in the past or the maximum value of an allowable range of the actual measurement value. Minimum value **1605** is the minimum value of the actual measurement value of each facility. In one aspect, minimum value **1605** may be the minimum value of the actual measurement value measured in the past or the minimum value of an allowable range of the actual measurement value.

(112) FIG. **17** is a view illustrating an example of a configuration of a setting file **1700** included in the editing data of the single-line diagram. In the example of FIG. **17**, setting file **1700** includes a setting file **1710** of the common setting and a setting file **1720** of the individual setting of monitoring device A.

(113) Setting file **1710** includes “layer 1” common to all the monitoring devices and a “symbol 1” and a “symbol 2” on “layer 1”. Furthermore, setting file **1710** includes the reference information about the internal data associated with each symbol and screen display information. For example, the “screen display information” includes information such as setting of the overlapping order and setting of display/non-display.

(114) Setting file **1720** includes “layer 2” for monitoring device A and a “symbol 10” on “layer 2”. Furthermore, similarly to setting file **1710**, setting file **1720** can include the reference information about the internal data associated with each symbol and the screen display information. As described above, setting file **1700** is a set of setting files managed for each layer.

(115) With reference to FIGS. **18** to **20**, the editing screen provided by editing portion **1401** of setting device **1400** will be described below. Editing portion **801** of setting device **400** can also provide the editing screens in FIGS. **18** to **20** to the user.

(116) FIG. **18** is a view illustrating a first example of a screen of an editing screen **1810** provided

by editing portion **1401**. Setting device **1400** can display editing screen **1810** on the display connected to output interface **906**. In addition, setting device **1400** can receive the operation input of the user to editing screen **1810** through various input instruments (such as a keyboard and/or a mouse) connected to input interface **905**. Furthermore, setting device **1400** can store the editing data of the single-line diagram edited on editing screen **1810** in secondary storage device **903** based on the reception of the storage operation input from the user.

(117) Editing screen **1810** includes a work region **1820**, a layer property **1830**, and a layer setting dialog **1840**. Work region **1820** provides a layer editing function. The user can define the layer by disposing the symbol in the region (the region of the layer) displayed in the work region.

(118) Layer property **1830** can display the list of defined layers. Furthermore, the user can refer to the information about each layer from layer property **1830**. For example, by referring to layer property **1830**, the user can determine whether each layer is the layer of the common setting for all the monitoring devices or the layer of the individual setting for a certain monitoring device. Furthermore, layer property **1830** can provide the function for setting the overlapping order of each layer and the function for setting the display/non-display of each layer.

(119) Layer setting dialog **1840** provides the function for defining the application target information about the layer. For example, the user can set whether a certain layer is the layer of the common setting or the layer of the individual setting of a certain monitoring device on layer setting dialog **1840**. In one aspect, layer setting dialog **1840** can be displayed when new layer preparation is selected from a menu or the like of editing screen **1810**. In another aspect, when the layer is selected from layer property **1830**, layer setting dialog **1840** may be displayed together with the application target information about the selected layer. In another aspect, layer setting dialog **1840** may be displayed as a screen different from editing screen **1810**. Furthermore, in another aspect, layer setting dialog **1840** may be displayed as a part of editing screen **1810**.

(120) FIG. **19** is a view illustrating a second example of the screen of editing screen **1810** provided by editing portion **1401**. Editing screen **1810** further includes a symbol panel **1910**. Symbol panel **1910** displays the list of symbols that can be disposed in work region **1820**. The user can select a certain symbol from symbol panel **1910** to dispose the selected symbol in work region **1820**. For example, the user may drag and drop the symbol from symbol panel **1910** to dispose the symbol in work region **1820**.

(121) FIG. **20** is a view illustrating a third example of the screen of editing screen **1810** provided by editing portion **1401**. Editing screen **1810** further includes an internal data dialog **2010**. Internal data dialog **2010** provides a function for associating the actual measurement value of each facility included in the internal data with each symbol. In one aspect, internal data dialog **2010** may be displayed as a screen separate from editing screen **1810**. In another aspect, internal data dialog **2010** may be displayed as a part of editing screen **1810**.

(122) As described above, setting device **1400** of the second embodiment has the function for associating the internal data with the symbol of each layer in addition to the function for generating the editing data/setting data of the single-line diagram for each monitoring device by diverting the common setting common to each monitoring device. With this function, the user can prepare the setting data of the single-line diagram for each monitoring device by diverting not only the symbol but also the internal data common to each monitoring device.

(123) Furthermore, editing screen **1810** provides the function for visually defining the layer and associating the internal data with each symbol. Using editing screen **1810**, the user can superimpose and check the common setting and the individual setting even while editing each setting. As a result, the user can easily prepare the setting data of the single-line diagram for each monitoring device as assumed. The respective embodiments may be implemented in an appropriate combination.

(124) It should be considered that the disclosed embodiments are an example in all respects and not restrictive. The scope of the present disclosure is defined by not the description above, but the

claims, and it is intended that all modifications within the meaning and scope of the claims and their equivalents are included in the present invention. In addition, the disclosed contents described in the embodiment and the modification are intended to be implemented alone or in combination as much as possible.

REFERENCE SIGNS LIST

(125) **100, 200A, 200B, 300A, 300B, 601A, 601B**: single-line diagram, **110, 210A, 210B, 220A, 220B**: symbol, **120, 1310**: internal data, **301A, 301B, 410, 1000, 1110A, 1110B, 1500**: editing data of single-line diagram, **303A, 303B, 1530, 1530A, 1530B**: setting data of single-line diagram, **310, 411, 1010, 1320**: common setting, **320, 412, 412A, 412B, 1020A, 1020B, 1330, 1330A, 1330B**: individual setting, **400, 1400**: setting device, **415, 804, 1404**: setting data output portion, **420A**: DB of monitoring device A, **420B**: DB of monitoring device B, **420C**: DB of monitoring device C, **600A, 600B, 700, 1300, 1700, 1710, 1720**: setting file, **801, 1401**: editing portion, **802, 1402**: layer processing portion, **803, 1403**: editing data storage, **805, 1405**: setting data storage, **900**: information processing device, **901**: CPU, **902**: primary storage device, **903**: secondary storage device, **904**: external instrument interface, **905**: input interface, **906**: output interface, **907**: communication interface, **1120A, 1120B**: display, **1406**: internal data storage, **1540, 1550A, 1550B**: association information, **1600**: incidental information, **1601**: ID, **1602**: unit, **1603**: data type, **1604**: maximum value, **1605**: minimum value, **1810**: editing screen, **1820**: work region, **1830**: layer property, **1840**: layer setting dialog, **1910**: symbol panel, **2010**: internal data dialog

Claims

1. A device comprising: an input portion to receive an operation related to a single-line diagram; a controller to generate setting data of the single-line diagram based on the operation; and a storage to store the setting data, wherein the input portion receives first setting related to a first power system, second setting related to a second power system, and third setting common to the first and second power systems, and the controller generates setting data of a first single-line diagram including the first setting and the third setting, generates setting data of a second single-line diagram including the second setting and the third setting, and stores the setting data of the first single-line diagram and the setting data of the second single-line diagram in the storage.
2. The device according to claim 1, wherein the first setting includes a first layer related to the first power system, the second setting includes a second layer related to the second power system, the third setting includes a third layer common to the first and second power systems, each of the layers includes one or a plurality of symbols, and the symbol is a figure indicating an instrument of a monitoring target.
3. The device according to claim 2, wherein the input portion receives input changing overlapping order of each of the layers, and the controller determines the overlapping order of each of the layers in the setting data of the first and second single-line diagrams based on the input changing the overlapping order.
4. The device according to claim 2, wherein the input portion receives input setting of display/non-display of each of the layers, and the controller determines display setting of each of the layers based on each setting of the display/non-display for each of the layers.
5. The device according to claim 4, wherein the generating the setting data of the first and second single-line diagrams includes: generating editing data of the first and second single-line diagrams including information about a combination of the first setting, the second setting, and the third setting; and generating the setting data of each of the first and second single-line diagrams based on the editing data of each of the first and second single-line diagrams.
6. The device according to claim 4, wherein the first setting includes first internal data related to the first power system, the second setting includes second internal data related to the second power system, the third setting includes third internal data common to the first power system and the

second power system, and each of the internal data is measured from each of a plurality of the instruments of the monitoring targets.

7. The device according to claim 6, wherein each of the internal data is a voltage value or a current value.

8. The device according to claim 6, wherein each of the internal data is associated with one of the symbols.

9. The device according to claim 8, wherein the generating the setting data of the first and second single-line diagrams includes: generating editing data of first and second single-line diagrams including information about a combination of each of the settings and information about association between each of the internal data and each of the symbols; and generating setting data for each of the first and second single-line diagrams based on the editing data for each of the first and second single-line diagrams.

10. A computer-implemented method for generating a single-line diagram comprising: receiving first setting related to a first power system, second setting related to a second power system, and third setting common to the first and second power systems; generating setting data of a first single-line diagram including the first setting and the third setting; generating setting data of a second single-line diagram including the second setting and the third setting; and storing the setting data of the first single-line diagram and the setting data of the second single-line diagram in a storage.

11. The computer-implemented method according to claim 10, wherein the first setting includes a first layer related to the first power system, the second setting includes a second layer related to the second power system, the third setting includes a third layer common to the first and second power systems, each of the layers includes one or a plurality of symbols, and the symbol is a figure indicating an instrument of a monitoring target.

12. The computer-implemented method according to claim 11, further comprising: receiving input changing overlapping order of each of the layers; and determining the overlapping order of each of the layers in setting data of the first and second single-line diagrams based on the input changing the overlapping order.

13. The computer-implemented method according to claim 11, further comprising: receiving input of setting of display/non-display of each of the layers; and determining a display setting for each of the layers based on each setting of the display/non-display for each of the layers.

14. The computer-implemented method according to claim 13, wherein the generating the setting data of the first and second single-line diagrams includes: generating editing data of the first and second single-line diagrams including information about a combination of the first setting, the second setting, and the third setting; and generating the setting data of each of the first and second single-line diagrams based on the editing data of each of the first and second single-line diagrams.

15. The computer-implemented method according to claim 13, wherein the first setting includes first internal data related to the first power system, the second setting includes second internal data related to the second power system, the third setting includes third internal data common to the first power system and the second power system, and each of the internal data is measured from each of a plurality of the instruments of the monitoring targets.

16. The computer-implemented method according to claim 15, wherein each of the internal data is a voltage value or a current value.

17. The computer-implemented method according to claim 15, wherein each of the internal data is associated with one of the symbols.

18. The computer-implemented method according to claim 17, wherein the generating the setting data of the first and second single-line diagrams includes: generating editing data of first and second single-line diagrams including information about a combination of each of the settings and information about association between each of the internal data and each of the symbols; and generating setting data for each of the first and second single-line diagrams based on the editing

data for each of the first and second single-line diagrams.

19. A non-transitory computer-readable medium comprising a program for causing one or a plurality of processors to execute the method of claim 10.
